

# Internal Dependencies

## References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

## Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List\_all\_Java\_artifacts

Table 1a - Top 30 artifacts with the highest package count

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |
| 1 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 2 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 3 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 4 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 5 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 6 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 7 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 8 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |

Table 1b - Top 30 artifacts with the highest type count

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |
| 1 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 2 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 3 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 4 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 5 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 6 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 7 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 8 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |
| 1 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 2 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 3 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 4 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 5 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 6 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 7 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 8 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |

Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 1 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 2 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 3 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 4 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 5 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 6 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 7 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 8 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |

Table 1e - Top 30 artifacts with the lowest package count

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 1 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 2 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 3 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 4 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 5 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 6 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 7 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 8 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |

Table 1f - Top 30 artifacts with the lowest type count

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 1 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 2 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 3 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 4 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 5 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 6 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 7 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 8 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |

Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |
| 1 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 2 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 3 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 4 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 5 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 6 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 7 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 8 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |

Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

|   | artifactName                              | packages | types | incomingDependencies | outgoingDependencies |
|---|-------------------------------------------|----------|-------|----------------------|----------------------|
| 0 | axon-messaging-4.12.2.jar                 | 70       | 830   | 8                    | 0                    |
| 1 | axon-modelling-4.12.2.jar                 | 10       | 158   | 6                    | 1                    |
| 2 | axon-tracing-opentelemetry-4.12.2.jar     | 1        | 5     | 1                    | 1                    |
| 3 | axon-eventsourcing-4.12.2.jar             | 9        | 133   | 5                    | 2                    |
| 4 | axon-disruptor-4.12.2.jar                 | 1        | 22    | 1                    | 3                    |
| 5 | axon-test-4.12.2.jar                      | 8        | 87    | 1                    | 3                    |
| 6 | axon-configuration-4.12.2.jar             | 1        | 43    | 2                    | 4                    |
| 7 | axon-server-connector-4.12.2.jar          | 11       | 147   | 1                    | 4                    |
| 8 | axon-spring-boot-autoconfigure-4.12.2.jar | 10       | 91    | 0                    | 7                    |

## Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

## Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic\\_Dependencies](#)

### Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

|    | artifactName                 | packageName                                       | dependentArtifactName        | dependentPackageName                                      | forwardToBackwardBalance | numberForward | numberBackward |
|----|------------------------------|---------------------------------------------------|------------------------------|-----------------------------------------------------------|--------------------------|---------------|----------------|
| 0  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.tracing                                 | 0.900000                 | 19            | 1              |
| 1  | axon-messaging-4.12.2        | org.axonframework.queryhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging.responsetypes                 | 0.882353                 | 16            | 1              |
| 2  | axon-messaging-4.12.2        | org.axonframework.queryhandling                   | axon-messaging-4.12.2        | org.axonframework.tracing                                 | 0.875000                 | 15            | 1              |
| 3  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging                               | 0.857143                 | 39            | 3              |
| 4  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging.annotation                    | 0.840000                 | 23            | 2              |
| 5  | axon-messaging-4.12.2        | org.axonframework.deadline                        | axon-messaging-4.12.2        | org.axonframework.tracing                                 | 0.800000                 | 9             | 1              |
| 6  | axon-messaging-4.12.2        | org.axonframework.commandhandling                 | axon-messaging-4.12.2        | org.axonframework.tracing                                 | 0.777778                 | 8             | 1              |
| 7  | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing                   | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing.eventstore                | 0.777778                 | 16            | 2              |
| 8  | axon-messaging-4.12.2        | org.axonframework.commandhandling.callbacks       | axon-messaging-4.12.2        | org.axonframework.commandhandling                         | 0.733333                 | 13            | 2              |
| 9  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.serialization                           | 0.647059                 | 14            | 3              |
| 10 | axon-messaging-4.12.2        | org.axonframework.messaging.unitofwork            | axon-messaging-4.12.2        | org.axonframework.messaging                               | 0.647059                 | 14            | 3              |
| 11 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector                    | 0.615385                 | 21            | 5              |
| 12 | axon-messaging-4.12.2        | org.axonframework.eventhandling.async             | axon-messaging-4.12.2        | org.axonframework.eventhandling                           | 0.571429                 | 11            | 3              |
| 13 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector                    | 0.555556                 | 21            | 6              |
| 14 | axon-messaging-4.12.2        | org.axonframework.eventhandling.replay            | axon-messaging-4.12.2        | org.axonframework.eventhandling                           | 0.454545                 | 8             | 3              |
| 15 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.util       | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector                    | 0.428571                 | 5             | 2              |
| 16 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.util         | 0.333333                 | 2             | 1              |
| 17 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query.subscription | 0.333333                 | 4             | 2              |

|    | artifactName                 | packageName                                                 | dependentArtifactName        | dependentPackageName                                       | forwardToBackwardBalance | numberForward | numberBackward |
|----|------------------------------|-------------------------------------------------------------|------------------------------|------------------------------------------------------------|--------------------------|---------------|----------------|
| 18 | axon-messaging-4.12.2        | org.axonframework.serialization.upcasting.event             | axon-messaging-4.12.2        | org.axonframework.eventhandling                            | 0.333333                 | 6             | 3              |
| 19 | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing.eventstore.jdbc             | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing.eventstore.jdbc.statements | 0.317073                 | 27            | 14             |
| 20 | axon-messaging-4.12.2        | org.axonframework.eventhandling                             | axon-messaging-4.12.2        | org.axonframework.eventhandling.tokenstore                 | 0.285714                 | 9             | 5              |
| 21 | axon-modelling-4.12.2        | org.axonframework.modelling.command.inspection              | axon-modelling-4.12.2        | org.axonframework.modelling.command                        | 0.250000                 | 20            | 12             |
| 22 | axon-messaging-4.12.2        | org.axonframework.queryhandling.registration                | axon-messaging-4.12.2        | org.axonframework.queryhandling                            | 0.250000                 | 5             | 3              |
| 23 | axon-messaging-4.12.2        | org.axonframework.messaging                                 | axon-messaging-4.12.2        | org.axonframework.serialization                            | 0.238095                 | 13            | 8              |
| 24 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.command              | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector                     | 0.222222                 | 11            | 7              |
| 25 | axon-messaging-4.12.2        | org.axonframework.messaging.annotation                      | axon-messaging-4.12.2        | org.axonframework.messaging.interceptors                   | 0.200000                 | 3             | 2              |
| 26 | axon-modelling-4.12.2        | org.axonframework.modelling.saga                            | axon-modelling-4.12.2        | org.axonframework.modelling.saga.metamodel                 | 0.142857                 | 4             | 3              |
| 27 | axon-messaging-4.12.2        | org.axonframework.commandhandling.distributed.commandfilter | axon-messaging-4.12.2        | org.axonframework.commandhandling.distributed              | 0.076923                 | 7             | 6              |

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown

Columns in addition to Table 2a:

- *dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

|    | artifactName          | packageName                     | dependentArtifactName | dependentPackageName                      | dependency                                                  | forwardToBackwardBalance | numberForward | numberBackwa |
|----|-----------------------|---------------------------------|-----------------------|-------------------------------------------|-------------------------------------------------------------|--------------------------|---------------|--------------|
| 0  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | EventBusSpanFactory->Span                                   | 0.900000                 | 19            |              |
| 1  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | EventProcessorSpanFactory->Span                             | 0.900000                 | 19            |              |
| 2  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventBusSpanFactory->Span                            | 0.900000                 | 19            |              |
| 3  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventProcessorSpanFactory->NoOpSpanFactory\$NoOpSpan | 0.900000                 | 19            |              |
| 4  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | EventMessage<-NestingSpanFactory                            | 0.900000                 | 19            |              |
| 5  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventProcessor\$Builder->SpanFactory                | 0.900000                 | 19            |              |
| 6  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventProcessorSpanFactory->SpanFactory               | 0.900000                 | 19            |              |
| 7  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventBusSpanFactory\$Builder->SpanFactory            | 0.900000                 | 19            |              |
| 8  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | SubscribingEventProcessor\$Builder->SpanFactory             | 0.900000                 | 19            |              |
| 9  | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventProcessorSpanFactory\$Builder->SpanFactory      | 0.900000                 | 19            |              |
| 10 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventBus->SpanScope                                 | 0.900000                 | 19            |              |
| 11 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventBusSpanFactory->SpanFactory                     | 0.900000                 | 19            |              |
| 12 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | SimpleEventBus\$Builder->SpanFactory                        | 0.900000                 | 19            |              |
| 13 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | TrackingEventProcessor\$Builder->SpanFactory                | 0.900000                 | 19            |              |
| 14 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventBus->Span                                      | 0.900000                 | 19            |              |
| 15 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultEventProcessorSpanFactory->Span                      | 0.900000                 | 19            |              |
| 16 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventProcessor\$Builder->NoOpSpanFactory            | 0.900000                 | 19            |              |
| 17 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventProcessor->Span                                | 0.900000                 | 19            |              |
| 18 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventBus\$Builder->NoOpSpanFactory                  | 0.900000                 | 19            |              |
| 19 | axon-messaging-4.12.2 | org.axonframework.eventhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | AbstractEventBus\$Builder->SpanFactory                      | 0.900000                 | 19            |              |
| 20 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | GenericSubscriptionQueryMessage->ResponseType               | 0.882353                 | 16            |              |
| 21 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | QuerySubscription->ResponseType                             | 0.882353                 | 16            |              |
| 22 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SimpleQueryUpdateEmitter->PublisherResponseType             | 0.882353                 | 16            |              |
| 23 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SimpleQueryBus->ResponseType                                | 0.882353                 | 16            |              |
| 24 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | QueryMessage->ResponseType                                  | 0.882353                 | 16            |              |
| 25 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SimpleQueryUpdateEmitter->MultipleInstancesResponseType     | 0.882353                 | 16            |              |
| 26 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | GenericStreamingQueryMessage->PublisherResponseType         | 0.882353                 | 16            |              |
| 27 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SimpleQueryUpdateEmitter->ResponseType                      | 0.882353                 | 16            |              |
| 28 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | QueryResponseMessage<-ConvertingResponseMessage             | 0.882353                 | 16            |              |
| 29 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | DefaultQueryGateway->ResponseType                           | 0.882353                 | 16            |              |
| 30 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | QueryGateway->ResponseType                                  | 0.882353                 | 16            |              |
| 31 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | GenericStreamingQueryMessage->ResponseType                  | 0.882353                 | 16            |              |
| 32 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SimpleQueryUpdateEmitter->OptionalResponseType              | 0.882353                 | 16            |              |
| 33 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | QueryGateway->ResponseTypes                                 | 0.882353                 | 16            |              |
| 34 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | GenericQueryMessage->ResponseType                           | 0.882353                 | 16            |              |
| 35 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | SubscriptionQueryMessage->ResponseType                      | 0.882353                 | 16            |              |
| 36 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.messaging.responsetypes | StreamingQueryMessage->ResponseType                         | 0.882353                 | 16            |              |
| 37 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultQueryBusSpanFactory\$Builder->SpanFactory            | 0.875000                 | 15            |              |
| 38 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultQueryUpdateEmitterSpanFactory->Span                  | 0.875000                 | 15            |              |
| 39 | axon-messaging-4.12.2 | org.axonframework.queryhandling | axon-messaging-4.12.2 | org.axonframework.tracing                 | DefaultQueryUpdateEmitterSpanFactory\$Builder->SpanFactory  | 0.875000                 | 15            |              |

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown\_BackwardOnly

|    | artifactName                 | packageName                                       | dependentArtifactName        | dependentPackageName                              | dependency                                                                                | forwardToBackward |
|----|------------------------------|---------------------------------------------------|------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------|
| 0  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.tracing                         | EventMessage<-NestingSpanFactory                                                          |                   |
| 1  | axon-messaging-4.12.2        | org.axonframework.queryhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging.responsetypes         | QueryResponseMessage<-ConvertingResponseMessage                                           |                   |
| 2  | axon-messaging-4.12.2        | org.axonframework.queryhandling                   | axon-messaging-4.12.2        | org.axonframework.tracing                         | QueryMessage<-SpanUtils                                                                   |                   |
| 3  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging                       | DomainEventMessage<-Headers                                                               |                   |
| 4  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging                       | TrackingToken<-StreamableMessageSource                                                    |                   |
| 5  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging                       | EventMessage<-Headers                                                                     |                   |
| 6  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging.annotation            | DomainEventMessage<-AggregateTypeParameterResolverFactory\$AggregateTypeParameterResolver |                   |
| 7  | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.messaging.annotation            | DomainEventMessage<-SourceIdParameterResolverFactory\$SourceIdParameterResolver           |                   |
| 8  | axon-messaging-4.12.2        | org.axonframework.deadline                        | axon-messaging-4.12.2        | org.axonframework.tracing                         | DeadlineMessage<-SpanUtils                                                                |                   |
| 9  | axon-messaging-4.12.2        | org.axonframework.commandhandling                 | axon-messaging-4.12.2        | org.axonframework.tracing                         | CommandMessage<-SpanUtils                                                                 |                   |
| 10 | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing                   | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing.eventstore        | EventStreamUtils<-DomainEventStream                                                       |                   |
| 11 | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing                   | axon-eventsourcing-4.12.2    | org.axonframework.eventsourcing.eventstore        | EventStreamUtils<-AbstractEventStorageEngine                                              |                   |
| 12 | axon-messaging-4.12.2        | org.axonframework.commandhandling.callbacks       | axon-messaging-4.12.2        | org.axonframework.commandhandling                 | LoggingCallback<-SimpleCommandBus\$Builder                                                |                   |
| 13 | axon-messaging-4.12.2        | org.axonframework.commandhandling.callbacks       | axon-messaging-4.12.2        | org.axonframework.commandhandling                 | NoOpCallback<-SimpleCommandBus\$Builder                                                   |                   |
| 14 | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.serialization                   | GapAwareTrackingToken<-GapAwareTrackingTokenConverter                                     |                   |
| 15 | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.serialization                   | GenericEventMessage<-AbstractXStreamSerializer                                            |                   |
| 16 | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | axon-messaging-4.12.2        | org.axonframework.serialization                   | GenericDomainEventMessage<-AbstractXStreamSerializer                                      |                   |
| 17 | axon-messaging-4.12.2        | org.axonframework.messaging.unitofwork            | axon-messaging-4.12.2        | org.axonframework.messaging                       | UnitOfWork<-DefaultInterceptorChain                                                       |                   |
| 18 | axon-messaging-4.12.2        | org.axonframework.messaging.unitofwork            | axon-messaging-4.12.2        | org.axonframework.messaging                       | UnitOfWork<-MessageHandlerInterceptor                                                     |                   |
| 19 | axon-messaging-4.12.2        | org.axonframework.messaging.unitofwork            | axon-messaging-4.12.2        | org.axonframework.messaging                       | CurrentUnitOfWork<-GenericMessage                                                         |                   |
| 20 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerEventStoreFactory\$Builder<-ServerConnectorConfigurerModule                     |                   |
| 21 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerEventStore\$Builder<-ServerConnectorConfigurerModule                            |                   |
| 22 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | EventProcessorInfoConfiguration<-ServerConnectorConfigurerModule                          |                   |
| 23 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerEventStore<-ServerConnectorConfigurerModule                                     |                   |
| 24 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.axon | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerEventStoreFactory<-ServerConnectorConfigurerModule                              |                   |
| 25 | axon-messaging-4.12.2        | org.axonframework.eventhandling.async             | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | SequencingPolicy<-SimpleEventHandlerInvoker                                               |                   |
| 26 | axon-messaging-4.12.2        | org.axonframework.eventhandling.async             | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | SequentialPerAggregatePolicy<-SimpleEventHandlerInvoker\$Builder                          |                   |
| 27 | axon-messaging-4.12.2        | org.axonframework.eventhandling.async             | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | SequencingPolicy<-SimpleEventHandlerInvoker\$Builder                                      |                   |
| 28 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode                             |                   |
| 29 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerQueryDispatchException<-ErrorCode                                               |                   |
| 30 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerQueryBus\$Builder<-ServerConnectorConfigurerModule                              |                   |
| 31 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | QueryPriorityCalculator<-ServerConnectorConfigurerModule                                  |                   |
| 32 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerQueryBus<-ServerConnectorConfigurerModule                                       |                   |
| 33 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.query      | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | AxonServerRemoteQueryHandlingException<-ErrorCode                                         |                   |
| 34 | axon-messaging-4.12.2        | org.axonframework.eventhandling.replay            | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | GenericResetContext<-AnnotationEventHandlerAdapter                                        |                   |
| 35 | axon-messaging-4.12.2        | org.axonframework.eventhandling.replay            | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | ResetContext<-ResetHandler                                                                |                   |
| 36 | axon-messaging-4.12.2        | org.axonframework.eventhandling.replay            | axon-messaging-4.12.2        | org.axonframework.eventhandling                   | ResetContext<-AnnotationEventHandlerAdapter                                               |                   |
| 37 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.util       | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | GrpcMessageSizeInterceptor<-AxonServerConnectionManager\$Builder                          |                   |
| 38 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.util       | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | ExceptionSerializer<-ErrorCode                                                            |                   |
| 39 | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector            | axon-server-connector-4.12.2 | org.axonframework.axonserver.connector.event.util | ErrorCode<-GrpcExceptionParser                                                            |                   |



# Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

## How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

## Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

|    | fullDependentTypeName                                                | declaredMethods | calledMethodNames                                                | calledMethods | callerTypes |
|----|----------------------------------------------------------------------|-----------------|------------------------------------------------------------------|---------------|-------------|
| 0  | org.axonframework.commandhandling.CommandMessage                     | 9               | [getCommandName]                                                 | 1             | 20          |
| 1  | org.axonframework.eventhandling.EventMessage                         | 9               | [getIdentifier]                                                  | 1             | 10          |
| 2  | org.axonframework.eventhandling.EventMessage                         | 9               | [getTimestamp, getIdentifier]                                    | 2             | 10          |
| 3  | org.axonframework.eventhandling.DomainEventMessage                   | 10              | [getSequenceNumber]                                              | 1             | 9           |
| 4  | org.axonframework.eventhandling.TrackedEventMessage                  | 10              | [trackingToken]                                                  | 1             | 8           |
| 5  | org.axonframework.commandhandling.GenericCommandResultMessage        | 14              | [asCommandResultMessage]                                         | 1             | 6           |
| 6  | org.axonframework.eventhandling.DomainEventMessage                   | 10              | [getSequenceNumber, getAggregateIdentifier, getType]             | 3             | 6           |
| 7  | org.axonframework.messaging.ResultMessage                            | 9               | [exceptionResult, isExceptional]                                 | 2             | 6           |
| 8  | org.axonframework.eventhandling.ReplayToken                          | 13              | [createReplayToken]                                              | 1             | 5           |
| 9  | org.axonframework.deadline.GenericDeadlineMessage                    | 11              | [asDeadlineMessage]                                              | 1             | 5           |
| 10 | org.axonframework.eventhandling.DomainEventMessage                   | 10              | [getAggregateIdentifier]                                         | 1             | 5           |
| 11 | org.axonframework.eventhandling.TrackedEventMessage                  | 12              | [trackingToken]                                                  | 1             | 4           |
| 12 | org.axonframework.eventhandling.GenericEventMessage                  | 11              | [asEventMessage]                                                 | 1             | 4           |
| 13 | org.axonframework.deadline.DeadlineMessage                           | 10              | [getDeadlineName]                                                | 1             | 4           |
| 14 | org.axonframework.eventhandling.DomainEventMessage                   | 10              | [getType]                                                        | 1             | 4           |
| 15 | org.axonframework.deadline.DefaultDeadlineManagerSpanFactory         | 8               | [builder]                                                        | 1             | 4           |
| 16 | org.axonframework.common.transaction.NoTransactionManager            | 4               | [instance]                                                       | 1             | 4           |
| 17 | org.axonframework.queryhandling.SimpleQueryUpdateEmitter             | 17              | [builder]                                                        | 1             | 3           |
| 18 | org.axonframework.commandhandling.GenericCommandResultMessage        | 15              | [asCommandResultMessage]                                         | 1             | 3           |
| 19 | org.axonframework.messaging.annotation.WrappedMessageHandlingMember  | 14              | [handle]                                                         | 1             | 3           |
| 20 | org.axonframework.eventhandling.ReplayToken                          | 13              | [isReplay]                                                       | 1             | 3           |
| 21 | org.axonframework.modelling.command.inspection.AggregateModel        | 13              | [type]                                                           | 1             | 3           |
| 22 | org.axonframework.queryhandling.SubscriptionQueryMessage             | 12              | [getUpdateResponseType]                                          | 1             | 3           |
| 23 | org.axonframework.eventhandling.DomainEventMessage                   | 11              | [getSequenceNumber, getAggregateIdentifier, getType]             | 3             | 3           |
| 24 | org.axonframework.queryhandling.SimpleQueryBus                       | 11              | [builder]                                                        | 1             | 3           |
| 25 | org.axonframework.eventhandling.DomainEventMessage                   | 10              | [getSequenceNumber, getAggregateIdentifier]                      | 2             | 3           |
| 26 | org.axonframework.eventhandling.GapAwareTrackingToken                | 10              | [advanceTo, newInstance, getIndex, withGapsTruncatedAt, getGaps] | 5             | 3           |
| 27 | org.axonframework.eventhandling.GenericEventMessage                  | 10              | [asEventMessage]                                                 | 1             | 3           |
| 28 | org.axonframework.queryhandling.DefaultQueryBusSpanFactory           | 10              | [builder]                                                        | 1             | 3           |
| 29 | org.axonframework.commandhandling.gateway.DefaultCommandGateway      | 9               | [builder]                                                        | 1             | 3           |
| 30 | org.axonframework.config.Configuration                               | 9               | [getComponent]                                                   | 1             | 3           |
| 31 | org.axonframework.config.Configuration                               | 9               | [upcasterChain, snapshotFilter]                                  | 2             | 3           |
| 32 | org.axonframework.eventhandling.EventMessage                         | 9               | [getTimestamp]                                                   | 1             | 3           |
| 33 | org.axonframework.messaging.MessageDecorator                         | 9               | [describeTo]                                                     | 1             | 3           |
| 34 | org.axonframework.queryhandling.QueryMessage                         | 9               | [getResponseType, getQueryName]                                  | 2             | 3           |
| 35 | org.axonframework.commandhandling.DefaultCommandBusSpanFactory       | 5               | [builder]                                                        | 1             | 3           |
| 36 | org.axonframework.eventhandling.TrackedEventData                     | 5               | [trackingToken]                                                  | 1             | 3           |
| 37 | org.axonframework.eventhandling.tokenstore.ConfigToken               | 5               | [get]                                                            | 1             | 3           |
| 38 | org.axonframework.eventhandling.DefaultEventBusSpanFactory           | 4               | [builder]                                                        | 1             | 3           |
| 39 | org.axonframework.queryhandling.DefaultQueryUpdateEmitterSpanFactory | 4               | [builder]                                                        | 1             | 3           |

## Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

|    | fullQualifiedDependentTypeName                                  | dependentTypeName            | dependentTypeLabels                                                                                                                                                                                                                                                                                                                                                                                                                                  | numberOfUsingPackages |
|----|-----------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 0  | org.axonframework.common.BuilderUtils                           | BuilderUtils                 | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Ma...                                                                                                                                          | 52                    |
| 1  | org.axonframework.common.AxonConfigurationException             | AxonConfigurationException   | [Type, File, Java, Class, ByteCode, Throwable, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedCom...                                                                                                                                          | 43                    |
| 2  | org.axonframework.messaging.Message                             | Message                      | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInduce...                                                                                                                                                                         | 42                    |
| 3  | org.axonframework.messaging.MetaData                            | MetaData                     | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4Ty...                                                                                                                                          | 40                    |
| 4  | org.axonframework.serialization.Serializer                      | Serializer                   | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...                                                                                                                                          | 37                    |
| 5  | org.axonframework.eventhandling.EventMessage                    | EventMessage                 | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeakl...                                                                                                                                          | 36                    |
| 6  | org.axonframework.messaging.unitofwork.UnitOfWork               | UnitOfWork                   | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWe...                                                                                                                                          | 33                    |
| 7  | org.axonframework.common.transaction.TransactionManager         | TransactionManager           | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1...                                                                                                                                          | 31                    |
| 8  | org.axonframework.common.Assert                                 | Assert                       | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Ma...                                                                                                                                          | 29                    |
| 9  | org.axonframework.serialization.SerializedObject                | SerializedObject             | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInduce...                                                                                                                                          | 28                    |
| 10 | org.axonframework.serialization.SerializedType                  | SerializedType               | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0]...                                                                                                                                         | 28                    |
| 11 | org.axonframework.messaging.unitofwork.CurrentUnitOfWork        | CurrentUnitOfWork            | [Type, File, Java, Class, ByteCode, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut21... | 22                    |
| 12 | org.axonframework.common.Registration                           | Registration                 | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity8, M...                                                                                                                                          | 22                    |
| 13 | org.axonframework.tracing.SpanFactory                           | SpanFactory                  | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity12,...                                                                                 | 22                    |
| 14 | org.axonframework.lifecycle.Lifecycle                           | Lifecycle                    | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut14, Mark4TypeHDBSCAN-1]                                                                                                                                                                                                            | 21                    |
| 15 | org.axonframework.lifecycle.Lifecycle\$LifecycleRegistry        | Lifecycle\$LifecycleRegistry | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut25, Mark4TypeHDBSCAN-1]                                                                                                                                                                                                            | 21                    |
| 16 | org.axonframework.messaging.MessageHandlerInterceptor           | MessageHandlerInterceptor    | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity6, Mark4Ty...                                                                                 | 20                    |
| 17 | org.axonframework.common.ObjectUtils                            | ObjectUtils                  | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut27, Mark4TypeHDBSCAN151]                                                                                                                                                                                                               | 20                    |
| 18 | org.axonframework.eventhandling.TrackingToken                   | TrackingToken                | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0]...                                                                                                                                                                     | 20                    |
| 19 | org.axonframework.eventhandling.DomainEventMessage              | DomainEventMessage           | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4Ty...                                                                                                                                          | 19                    |
| 20 | org.axonframework.messaging.MessageDispatchInterceptor          | MessageDispatchInterceptor   | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity7, Mark4TypeLeidenCommunity8, Mark4TypePec...                     | 19                    |
| 21 | org.axonframework.messaging.annotation.ParameterResolverFactory | ParameterResolverFactory     | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, ...                                                                                                                                          | 19                    |
| 22 | org.axonframework.serialization.SimpleSerializedObject          | SimpleSerializedObject       | [Type, File, Java, Class, GenericDeclaration, ByteCode, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity5, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut18, Mark4Type...                                                                                                                                          | 19                    |
| 23 | org.axonframework.common.AxonNonTransientException              | AxonNonTransientException    | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecompositi...                                                                                                                                          | 18                    |
| 24 | org.axonframework.eventhandling.GenericEventMessage             | GenericEventMessage          | [Type, File, Java, Class, GenericDeclaration, ByteCode, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity5, Mark4TypeKCoreD...                                                                                                                                          | 18                    |
| 25 | org.axonframework.commandhandling.CommandMessage                | CommandMessage               | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeakl...                                                                                                                                          | 17                    |
| 26 | org.axonframework.tracing.NoOpSpanFactory                       | NoOpSpanFactory              | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity12, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut56, Mark4TypeHDBSCAN111]                                                                                                                                                                                                              | 17                    |
| 27 | org.axonframework.eventhandling.EventBus                        | EventBus                     | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, Mark4TypeKCoreDecomposition10, ...                                                                                                                                          | 15                    |
| 28 | org.axonframework.messaging.annotation.HandlerDefinition        | HandlerDefinition            | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition10, ...                                                                                                                                          | 15                    |
| 29 | org.axonframework.common.transaction.NoTransactionManager       | NoTransactionManager         | [Type, File, Java, ByteCode, Enum, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity6, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCut65, Mark4TypeLocalClusteringCoefficient0.14285714285714285, Mark4TypeHDBSCAN207]                                                                                                                                                        | 15                    |
| 30 | org.axonframework.messaging.ResultMessage                       | ResultMessage                | [Type, File, Java, GenericDeclaration, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity...                                                                                                                                                                         | 15                    |
| 31 | org.axonframework.tracing.Span                                  | Span                         | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4,...                                                                                                                                          | 15                    |
| 32 | org.axonframework.messaging.unitofwork.DefaultUnitOfWork        | DefaultUnitOfWork            | [Type, File, Java, Class, GenericDeclaration, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeKCoreDecomposition10, Mark4TypeMaximumKCuto, Mark4TypeHDBSCAN141]                                                                                                                                                                                            | 14                    |
| 33 | org.axonframework.common.ReflectionUtils                        | ReflectionUtils              | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity3, Mark4TypeLeidenCommunity3, Mark4TypeKCoreDecomposition7,                                                                                                                                                                                                                                                             | 14                    |

|    | fullQualifiedDependentTypeName                            | dependentTypeName         | dependentTypeLabels                                                                                                                                                                                                                                                                                        | numberOfUsingPackages |
|----|-----------------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                           |                           | Mark4TypeMaximumKCut94, Mark4TypeHDBSCAN113, Mark4TopAnomalyHub]                                                                                                                                                                                                                                           |                       |
| 34 | org.axonframework.messaging.InterceptorChain              | InterceptorChain          | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut0, Mark4TypeLocalClusteringCoefficient0.1038961038961039, Mark4TypeHDBSCAN-1]             | 13                    |
| 35 | org.axonframework.eventhandling.TrackedEventMessage       | TrackedEventMessage       | Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity10, Mark4T...                                                                                           | 13                    |
| 36 | org.axonframework.common.AxonException                    | AxonException             | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation22, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity10, Mark4TypeCoreDecomposi... | 12                    |
| 37 | org.axonframework.common.DateTimeUtils                    | DateTimeUtils             | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity1, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut36, Mark4TypeHDBSCAN153]                                                                       | 12                    |
| 38 | org.axonframework.messaging.DefaultInterceptorChain       | DefaultInterceptorChain   | [Type, File, Java, Class, GenericDeclaration, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity8, Mark4TypeLeidenCommunity5, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut65, Mark4TypeLocalClusteringCoefficient0.25, Mark4TypeHDBSCAN-1]          | 12                    |
| 39 | org.axonframework.eventhandling.GenericDomainEventMessage | GenericDomainEventMessage | [Type, File, Java, Class, GenericDeclaration, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity5, Mark4TypeCoreDecomposition10, Mark4TypeMaximumKCut9, Mark4TypeLocalClusteringCoefficient0.2727272727272727, Mark4TypeHDB... | 12                    |

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report `ArtifactPackageUsage` .

|    | artifactName                          | dependentArtifactName        | dependentPackages | dependentArtifactPackages | packageUsagePercentage | dependentFullQualifiedPackageNames                                                                                                                                                                                                                                                                            | dependentPackageNames                                                                                                                                                                                                                                                                                        |
|----|---------------------------------------|------------------------------|-------------------|---------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | axon-tracing-opentelemetry-4.12.2     | axon-messaging-4.12.2        | 3                 | 70                        | 0.042857               | [org.axonframework.messaging, org.axonframework.common, org.axonframework.tracing]                                                                                                                                                                                                                            | [messaging, common, tracing]                                                                                                                                                                                                                                                                                 |
| 1  | axon-server-connector-4.12.2          | axon-modelling-4.12.2        | 1                 | 10                        | 0.100000               | [org.axonframework.modelling.command]                                                                                                                                                                                                                                                                         | [command]                                                                                                                                                                                                                                                                                                    |
| 2  | axon-spring-boot-autoconfigure-4.12.2 | axon-test-4.12.2             | 1                 | 8                         | 0.125000               | [org.axonframework.test.server]                                                                                                                                                                                                                                                                               | [server]                                                                                                                                                                                                                                                                                                     |
| 3  | axon-disruptor-4.12.2                 | axon-messaging-4.12.2        | 9                 | 70                        | 0.128571               | [org.axonframework.monitoring, org.axonframework.commandhandling.callbacks, org.axonframework.common, org.axonframework.messaging, org.axonframework.messaging.annotation, org.axonframework.common.transaction, org.axonframework.common.caching, org.axonframework.commandhandling, org.axonframework....]  | [monitoring, callbacks, common, messaging, annotation, transaction, caching, commandhandling, unitofwork]                                                                                                                                                                                                    |
| 4  | axon-test-4.12.2                      | axon-messaging-4.12.2        | 10                | 70                        | 0.142857               | [org.axonframework.eventhandling.scheduling, org.axonframework.common, org.axonframework.eventhandling, org.axonframework.commandhandling, org.axonframework.messaging, org.axonframework.deadline, org.axonframework.messaging.unitofwork, org.axonframework.commandhandling.gateway, org.axonframework...]  | [scheduling, common, eventhandling, commandhandling, messaging, deadline, unitofwork, gateway, annotation, stream]                                                                                                                                                                                           |
| 5  | axon-disruptor-4.12.2                 | axon-modelling-4.12.2        | 2                 | 10                        | 0.200000               | [org.axonframework.modelling.command.inspection, org.axonframework.modelling.command]                                                                                                                                                                                                                         | [inspection, command]                                                                                                                                                                                                                                                                                        |
| 6  | axon-eventsourcing-4.12.2             | axon-modelling-4.12.2        | 2                 | 10                        | 0.200000               | [org.axonframework.modelling.command, org.axonframework.modelling.command.inspection]                                                                                                                                                                                                                         | [command, inspection]                                                                                                                                                                                                                                                                                        |
| 7  | axon-test-4.12.2                      | axon-eventsourcing-4.12.2    | 2                 | 9                         | 0.222222               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing]                                                                                                                                                                                                                                 | [eventstore, eventsourcing]                                                                                                                                                                                                                                                                                  |
| 8  | axon-disruptor-4.12.2                 | axon-eventsourcing-4.12.2    | 2                 | 9                         | 0.222222               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing]                                                                                                                                                                                                                                 | [eventstore, eventsourcing]                                                                                                                                                                                                                                                                                  |
| 9  | axon-modelling-4.12.2                 | axon-messaging-4.12.2        | 18                | 70                        | 0.257143               | [org.axonframework.common, org.axonframework.common.property, org.axonframework.messaging.annotation, org.axonframework.common.annotation, org.axonframework.commandhandling, org.axonframework.messaging.unitofwork, org.axonframework.eventhandling, org.axonframework.messaging, org.axonframework.de...]  | [common, property, annotation, commandhandling, unitofwork, eventhandling, messaging, deadline, jpa, lock, tracing, interceptors, serialization, xml, jdbc, legacyjpa, caching]                                                                                                                              |
| 10 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | 20                | 70                        | 0.285714               | [org.axonframework.eventhandling, org.axonframework.messaging.unitofwork, org.axonframework.common, org.axonframework.serialization, org.axonframework.common.jdbc, org.axonframework.common.transaction, org.axonframework.serialization.upcasting.event, org.axonframework.messaging, org.axonframework...] | [eventhandling, unitofwork, common, serialization, jdbc, transaction, event, messaging, commandhandling, annotation, io, jpa, lifecycle, stream, xml, monitoring, tracing, caching, lock, legacyjpa]                                                                                                         |
| 11 | axon-server-connector-4.12.2          | axon-eventsourcing-4.12.2    | 3                 | 9                         | 0.333333               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing, org.axonframework.eventsourcing.snapshotting]                                                                                                                                                                                   | [eventstore, eventsourcing, snapshotting]                                                                                                                                                                                                                                                                    |
| 12 | axon-server-connector-4.12.2          | axon-messaging-4.12.2        | 25                | 70                        | 0.357143               | [org.axonframework.commandhandling.distributed, org.axonframework.tracing, org.axonframework.commandhandling, org.axonframework.lifecycle, org.axonframework.serialization, org.axonframework.commandhandling.callbacks, org.axonframework.messaging, org.axonframework.common, org.axonframework.queryh...]  | [distributed, tracing, commandhandling, lifecycle, serialization, callbacks, messaging, common, queryhandling, util, responsesetups, tokenstore, pooled, transaction, eventhandling, inmemory, scheduling, event, async, monitoring, stream, unitofwork, jdbc, xml, java]                                    |
| 13 | axon-spring-boot-autoconfigure-4.12.2 | axon-server-connector-4.12.2 | 4                 | 11                        | 0.363636               | [org.axonframework.axonserver.connector, org.axonframework.axonserver.connector.query, org.axonframework.axonserver.connector.command, org.axonframework.axonserver.connector.event.axon]                                                                                                                     | [connector, query, command, axon]                                                                                                                                                                                                                                                                            |
| 14 | axon-configuration-4.12.2             | axon-eventsourcing-4.12.2    | 4                 | 9                         | 0.444444               | [org.axonframework.eventsourcing.snapshotting, org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.eventstore.jpa, org.axonframework.eventsourcing]                                                                                                                                   | [snapshotting, eventstore, jpa, eventsourcing]                                                                                                                                                                                                                                                               |
| 15 | axon-test-4.12.2                      | axon-modelling-4.12.2        | 5                 | 10                        | 0.500000               | [org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository.inmemory, org.axonframework.modelling.saga.repository, org.axonframework.modelling.command, org.axonframework.modelling.command.inspection]                                                                                    | [saga, inmemory, repository, command, inspection]                                                                                                                                                                                                                                                            |
| 16 | axon-configuration-4.12.2             | axon-messaging-4.12.2        | 37                | 70                        | 0.528571               | [org.axonframework.messaging.unitofwork, org.axonframework.common.jdbc, org.axonframework.common.jpa, org.axonframework.eventhandling.deadletter, org.axonframework.commandhandling, org.axonframework.tokenstore.inmemory, org.axonframework.deadline, org.axonframework.eventhandling.to...]                | [unitofwork, jdbc, jpa, deadletter, commandhandling, inmemory, deadline, monitoring, serialization, messaging, lifecycle, queryhandling, scheduling, common, transaction, tokenstore, async, correlation, detection, tracing, updates, annotation, util, interceptors, caching, configuration, gateway, ...] |
| 17 | axon-spring-boot-autoconfigure-4.12.2 | axon-eventsourcing-4.12.2    | 5                 | 9                         | 0.555556               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing, org.axonframework.eventsourcing.eventstore.jdbc, org.axonframework.eventsourcing.eventstore.jpa, org.axonframework.eventsourcing.eventstore.legacyjpa]                                                                          | [eventstore, eventsourcing, jdbc, jpa, legacyjpa]                                                                                                                                                                                                                                                            |
| 18 | axon-modelling-4.12.2                 | axon-modelling-4.12.2        | 6                 | 10                        | 0.600000               | [org.axonframework.modelling.command, org.axonframework.modelling.command.inspection, org.axonframework.modelling.saga.metamodel, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository.jpa]                                             | [command, inspection, metamodel, repository, saga, jpa]                                                                                                                                                                                                                                                      |
| 19 | axon-spring-boot-autoconfigure-4.12.2 | axon-modelling-4.12.2        | 6                 | 10                        | 0.600000               | [org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository, org.axonframework.modelling.saga.repository.jdbc, org.axonframework.modelling.command, org.axonframework.modelling.saga.repository.jpa, org.axonframework.modelling.saga.repository.legacyjpa]                                | [saga, repository, jdbc, command, jpa, legacyjpa]                                                                                                                                                                                                                                                            |
| 20 | axon-configuration-4.12.2             | axon-modelling-4.12.2        | 6                 | 10                        | 0.600000               | [org.axonframework.modelling.saga, org.axonframework.modelling.saga.repository.jpa, org.axonframework.modelling.command.inspection, org.axonframework.modelling.saga.repository, org.axonframework.modelling.command, org.axonframework.modelling.saga.repository.inmemory]                                   | [saga, jpa, inspection, repository, command, inmemory]                                                                                                                                                                                                                                                       |
| 21 | axon-test-4.12.2                      | axon-test-4.12.2             | 5                 | 8                         | 0.625000               | [org.axonframework.test, org.axonframework.test.matchers, org.axonframework.test.eventscheduler, org.axonframework.test.decline, org.axonframework.test.utils]                                                                                                                                                | [test, matchers, decline, eventscheduler, utils]                                                                                                                                                                                                                                                             |
| 22 | axon-spring-boot-autoconfigure-4.12.2 | axon-messaging-4.12.2        | 44                | 70                        | 0.628571               | [org.axonframework.common.jpa, org.axonframework.messaging.timeout, org.axonframework.common.legacyjpa, org.axonframework.common.jdbc, org.axonframework.serialization, org.axonframework.eventhandling.scheduling.dbscheduler, org.axonframework.eventhandling.deadletter.jdbc, org.axonframework.seria...]  | [jpa, timeout, legacyjpa, jdbc, serialization, dbscheduler, xml, commandhandling, common, jobrunr, configuration, updates, gateway, correlation, scheduling, queryhandling, lifecycle, detection, annotation, attributes, tokenstore, pooled, deadletter, messaging, avro, transaction, tracing, interce...] |
| 23 | axon-messaging-4.12.2                 | axon-messaging-4.12.2        | 44                | 70                        | 0.628571               | [org.axonframework.util, org.axonframework.serialization, org.axonframework.common, org.axonframework.queryhandling, org.axonframework.messaging, org.axonframework.commandhandling, org.axonframework.common.transaction, org.axonframework.common.annotation, org.axonframework.messaging.annotation, ...]  | [util, serialization, common, queryhandling, messaging, commandhandling, transaction, annotation, eventhandling, distributed, unitofwork, tokenstore, jdbc, deadletter, api, async, stream, tracing, io, monitoring, replay, event, lifecycle, jobrunr, scheduling, deadline, jpa, correlation, digest, ...] |
| 24 | axon-eventsourcing-4.12.2             | axon-eventsourcing-4.12.2    | 7                 | 9                         | 0.777778               | [org.axonframework.eventsourcing.eventstore, org.axonframework.eventsourcing.eventstore.jdbc.statements, org.axonframework.eventsourcing.snapshotting, org.axonframework.eventsourcing.eventstore.jdbc, org.axonframework.eventsourcing, org.axonframework.eventsourcing.conflictresolution, org.axonfra...]  | [eventstore, statements, snapshotting, jdbc, eventsourcing, conflictresolution, jpa]                                                                                                                                                                                                                         |

|    | artifactName                          | dependentArtifactName                 | dependentPackages | dependentArtifactPackages | packageUsagePercentage | dependentFullQualifiedPackageNames                                                                                                                                                                                                                                                                          | dependentPackageNames                                                                     |
|----|---------------------------------------|---------------------------------------|-------------------|---------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 25 | axon-server-connector-4.12.2          | axon-server-connector-4.12.2          | 9                 | 11                        | 0.818182               | [org.axonframework.axonserver.connector, org.axonframework.axonserver.connector.util, org.axonframework.axonserver.connector.command, org.axonframework.axonserver.connector.query.subscription, org.axonframework.axonserver.connector.event.util, org.axonframework.axonserver.connector.query, org.ax... | [connector, util, command, subscription, query, heartbeat, axon, processor]               |
| 26 | axon-spring-boot-autoconfigure-4.12.2 | axon-spring-boot-autoconfigure-4.12.2 | 9                 | 10                        | 0.900000               | [org.axonframework.actuator, org.axonframework.springboot.service.connection, org.axonframework.actuator.axonserver, org.axonframework.springboot.util, org.axonframework.springboot, org.axonframework.springboot.util.jpa, org.axonframework.springboot.updates, org.axonframework.springboot.autoconf... | [actuator, connection, axonserver, util, springboot, jpa, updates, autoconfig, legacyjpa] |
| 27 | axon-server-connector-4.12.2          | axon-configuration-4.12.2             | 1                 | 1                         | 1.000000               | [org.axonframework.config]                                                                                                                                                                                                                                                                                  | [config]                                                                                  |
| 28 | axon-configuration-4.12.2             | axon-disruptor-4.12.2                 | 1                 | 1                         | 1.000000               | [org.axonframework.disruptor.commandhandling]                                                                                                                                                                                                                                                               | [commandhandling]                                                                         |
| 29 | axon-spring-boot-autoconfigure-4.12.2 | axon-configuration-4.12.2             | 1                 | 1                         | 1.000000               | [org.axonframework.config]                                                                                                                                                                                                                                                                                  | [config]                                                                                  |

Table 7 - Types that are used by multiple artifacts

This table shows the top 30 types that only use a few (compared to all existing) types of another artifact. The whole table can be found in the CSV report `ClassesPerPackageUsageAcrossArtifacts` .

|    | artifactName                          | dependentArtifactName        | packageName                                                | dependentPackage.fqn                   | dependentTypes | dependentPackageTypes | typeUsagePercentage |                     |
|----|---------------------------------------|------------------------------|------------------------------------------------------------|----------------------------------------|----------------|-----------------------|---------------------|---------------------|
| 0  | axon-modelling-4.12.2                 | axon-messaging-4.12.2        | org.axonframework.modelling.saga.metamodel                 | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 1  | axon-spring-boot-autoconfigure-4.12.2 | axon-messaging-4.12.2        | org.axonframework.springboot.autoconfigure.legacyjpa       | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 2  | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.snapshottng                | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 3  | axon-test-4.12.2                      | axon-messaging-4.12.2        | org.axonframework.test.matchers                            | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 4  | axon-spring-boot-autoconfigure-4.12.2 | axon-messaging-4.12.2        | org.axonframework.springboot.util                          | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 5  | axon-server-connector-4.12.2          | axon-messaging-4.12.2        | org.axonframework.axonserver.connector                     | org.axonframework.eventhandling        | 1              | 100                   | 0.010000            |                     |
| 6  | axon-eventsourcing-4.12.2             | axon-modelling-4.12.2        | org.axonframework.eventsourcing.eventstore.jdbc            | org.axonframework.modelling.command    | 1              | 56                    | 0.017857            | [org.a              |
| 7  | axon-eventsourcing-4.12.2             | axon-modelling-4.12.2        | org.axonframework.eventsourcing.conflictresolution         | org.axonframework.modelling.command    | 1              | 56                    | 0.017857            | [org.axonframework. |
| 8  | axon-server-connector-4.12.2          | axon-modelling-4.12.2        | org.axonframework.axonserver.connector                     | org.axonframework.modelling.command    | 1              | 56                    | 0.017857            | [org.a              |
| 9  | axon-test-4.12.2                      | axon-messaging-4.12.2        | org.axonframework.test.eventscheduler                      | org.axonframework.eventhandling        | 2              | 100                   | 0.020000            |                     |
| 10 | axon-modelling-4.12.2                 | axon-messaging-4.12.2        | org.axonframework.modelling.command.legacyjpa              | org.axonframework.eventhandling        | 2              | 100                   | 0.020000            | [org.a              |
| 11 | axon-modelling-4.12.2                 | axon-messaging-4.12.2        | org.axonframework.modelling.command                        | org.axonframework.eventhandling        | 2              | 100                   | 0.020000            | [org.a              |
| 12 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.conflictresolution         | org.axonframework.eventhandling        | 2              | 100                   | 0.020000            |                     |
| 13 | axon-spring-boot-autoconfigure-4.12.2 | axon-configuration-4.12.2    | org.axonframework.springboot                               | org.axonframework.config               | 1              | 43                    | 0.023256            |                     |
| 14 | axon-spring-boot-autoconfigure-4.12.2 | axon-configuration-4.12.2    | org.axonframework.springboot.autoconfigure.legacyjpa       | org.axonframework.config               | 1              | 43                    | 0.023256            |                     |
| 15 | axon-server-connector-4.12.2          | axon-configuration-4.12.2    | org.axonframework.axonserver.connector.processor           | org.axonframework.config               | 1              | 43                    | 0.023256            |                     |
| 16 | axon-server-connector-4.12.2          | axon-eventsourcing-4.12.2    | org.axonframework.axonserver.connector.event.axon          | org.axonframework.eventsourcing        | 1              | 42                    | 0.023810            |                     |
| 17 | axon-test-4.12.2                      | axon-messaging-4.12.2        | org.axonframework.test.matchers                            | org.axonframework.messaging            | 1              | 35                    | 0.028571            |                     |
| 18 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.conflictresolution         | org.axonframework.messaging            | 1              | 35                    | 0.028571            |                     |
| 19 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.eventstore                 | org.axonframework.messaging            | 1              | 35                    | 0.028571            |                     |
| 20 | axon-test-4.12.2                      | axon-messaging-4.12.2        | org.axonframework.test                                     | org.axonframework.messaging            | 1              | 35                    | 0.028571            |                     |
| 21 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.eventstore.jdbc            | org.axonframework.serialization        | 1              | 34                    | 0.029412            |                     |
| 22 | axon-spring-boot-autoconfigure-4.12.2 | axon-messaging-4.12.2        | org.axonframework.springboot.autoconfigure.legacyjpa       | org.axonframework.serialization        | 1              | 34                    | 0.029412            |                     |
| 23 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.eventstore.jpa             | org.axonframework.serialization        | 1              | 34                    | 0.029412            |                     |
| 24 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.eventstore.legacyjpa       | org.axonframework.serialization        | 1              | 34                    | 0.029412            |                     |
| 25 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.eventstore.jdbc.statements | org.axonframework.eventhandling        | 3              | 100                   | 0.030000            | org.a               |
| 26 | axon-test-4.12.2                      | axon-modelling-4.12.2        | org.axonframework.test.utils                               | org.axonframework.modelling.saga       | 1              | 33                    | 0.030303            | [                   |
| 27 | axon-eventsourcing-4.12.2             | axon-messaging-4.12.2        | org.axonframework.eventsourcing.conflictresolution         | org.axonframework.commandhandling      | 1              | 32                    | 0.031250            |                     |
| 28 | axon-test-4.12.2                      | axon-messaging-4.12.2        | org.axonframework.test.matchers                            | org.axonframework.commandhandling      | 1              | 32                    | 0.031250            |                     |
| 29 | axon-spring-boot-autoconfigure-4.12.2 | axon-server-connector-4.12.2 | org.axonframework.springboot.service.connection            | org.axonframework.axonserver.connector | 1              | 32                    | 0.031250            | [org.axon           |

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

packageName duplicates artifactNames

Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.





